SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Fundamentals of Data Science **COURSE CODE: DSA04**

COURSE OUTCOME COVERED

CO1: *Analyse the role of data science, facets of data, and the big data ecosystem for informed decision-making. (BL4)*

QUESTIONS: 05 **Total Marks:100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I Jeffrey Samuels *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jeffrey Samuels

Annexure A

Concept Mapping

1. A college wants to analyze student performance using attendance records, internal marks, assignment scores, and final exam results. Draw a concept map showing the Data Science process from data collection to data transformation.
2. A hospital collects patient details from registration forms, lab reports, medical devices, and doctor notes. Construct a concept map showing the different data collection sources and how they are used for analysis.
3. An online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis.
4. A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.
5. A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.

RUBRICS FOR CALCULATION

Concept Mapping					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

CO1-ASSESSMENT TOOL -2 Concept Map

QUESTIONS : 5 **MARKS : 50**

Q No	Concept Identification (25%)	Linkages (25%)	Organisation (20%)	Integration of Literature (20%)	Clarity (10%)	Total Score (100%)
1	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
2	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
3	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
4	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
5	2.5 /2.5	2.5 /2.5	1 /2.0	1 /2.0	1 /1.0	8 /10
Overall Rubrics Value	12.5 /12.5	12.5 /12.5	5 /10	5 /10	5 /10	40 /50
	25 /25	25 /25	10 /20	10 /20	10 /10	80 /100

DSA0401 - Fundamentals of Data Science

Name :- E. Jeffrey Samuel
Reg. No :- 192524412
Dept :- B. Tech AIDS
Date :- 8/7/2026

A college wants to analyse the students performance using attendance records, internal marks, assignments, and final exam results. Draw a concept map showing the data science process from data collection to data Transformation.

40/50

Student Performance Analysis

Problem Definition
Analyse overall student academic performance

Attendance Records

Internal Marks

1. Data Collection

Assignment

Final Exam Result

2. Data Cleaning & Preprocessing
handle missing values, remove duplicates + fix formats.

3. Data Transformation
normalize score, compute weighted averge

4. Data Analysis & Exploration
statistics, correlation (attendance vs marks)

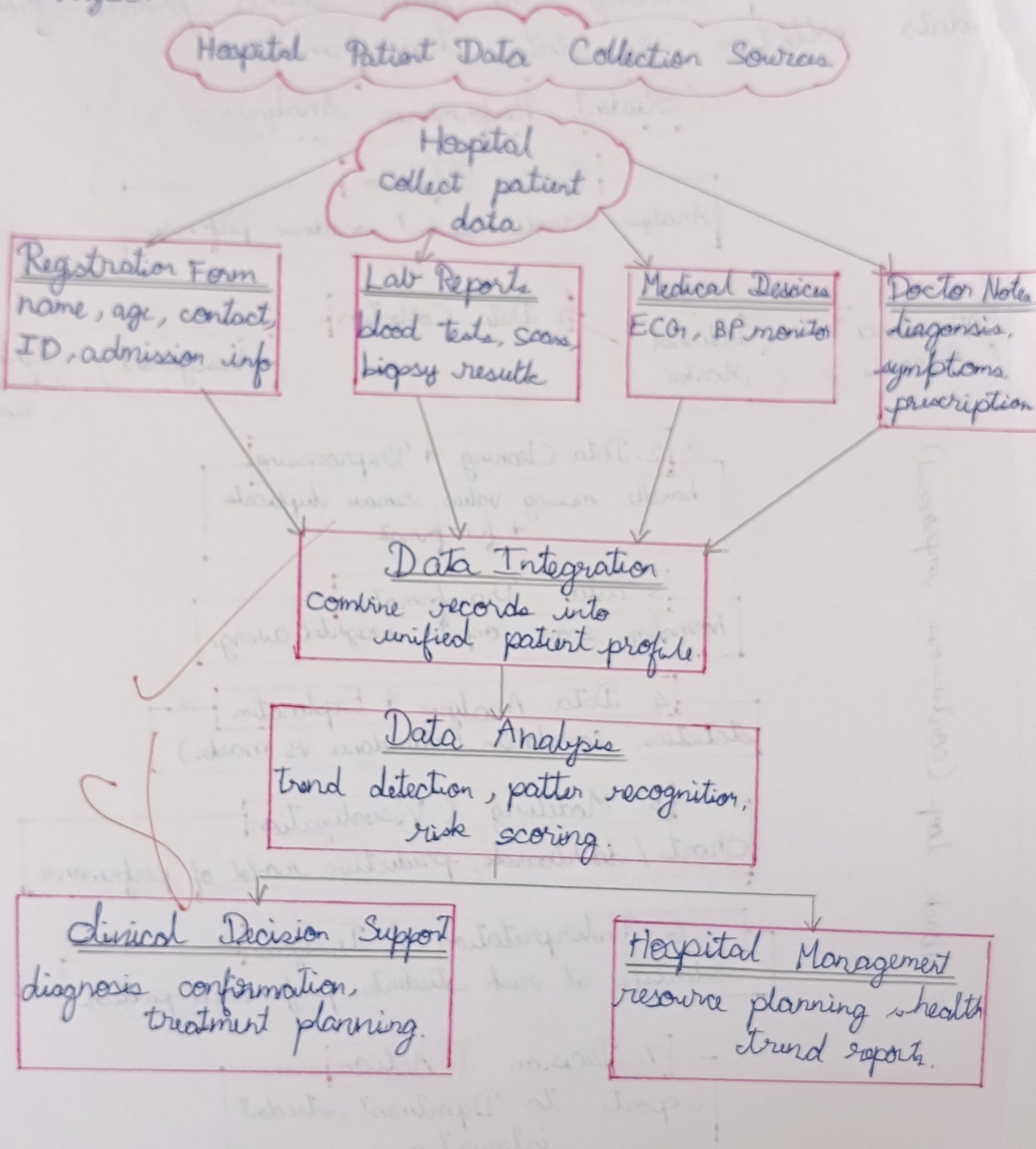
5. Modeling & Visualization
Charts / dashboards, predictive model of performance

6. Interpretation & Insights
identify at risk students, performance patterns

7. Decision & Action
reports to Department, student interventions.

Feedback loop (continuous improvement)

2. A Hospital collects a patient details through reg form, lab reports, medical devices and doctor notes. Construct a Concept map shows a different data collection sources and how They are used for analysis.



A online shopping company finds missing customer details, duplicate orders, wrong product entries, and unusual purchase amounts in its dataset. Draw a concept map showing the data cleaning steps required before analysis

Problem

- Missing customer details
- Duplicate orders
- Wrong product entries
- Unusual purchase amounts

Data Cleaning Steps Before Analysis

1. Identify Issues

- Scan the dataset
- Detect missing values
- Find duplicate records
- Detect unusual amounts

2. Handle Missing Customer Details.

- Check for null / blank values
- Fill missing values
- Or remove incomplete records if necessary.

3. Remove Duplicate Orders.

- Identify duplicates using Order ID, Customer ID, Date, Amount.
- Keep the first valid record and remove others.

4. Correct Wrong Product Entries.

- Validate product names/IDs with product master list.
- Correct spelling or mapping errors.
- Remove invalid or unknown products

5. Detect & Treat Unusual Purchase Amounts

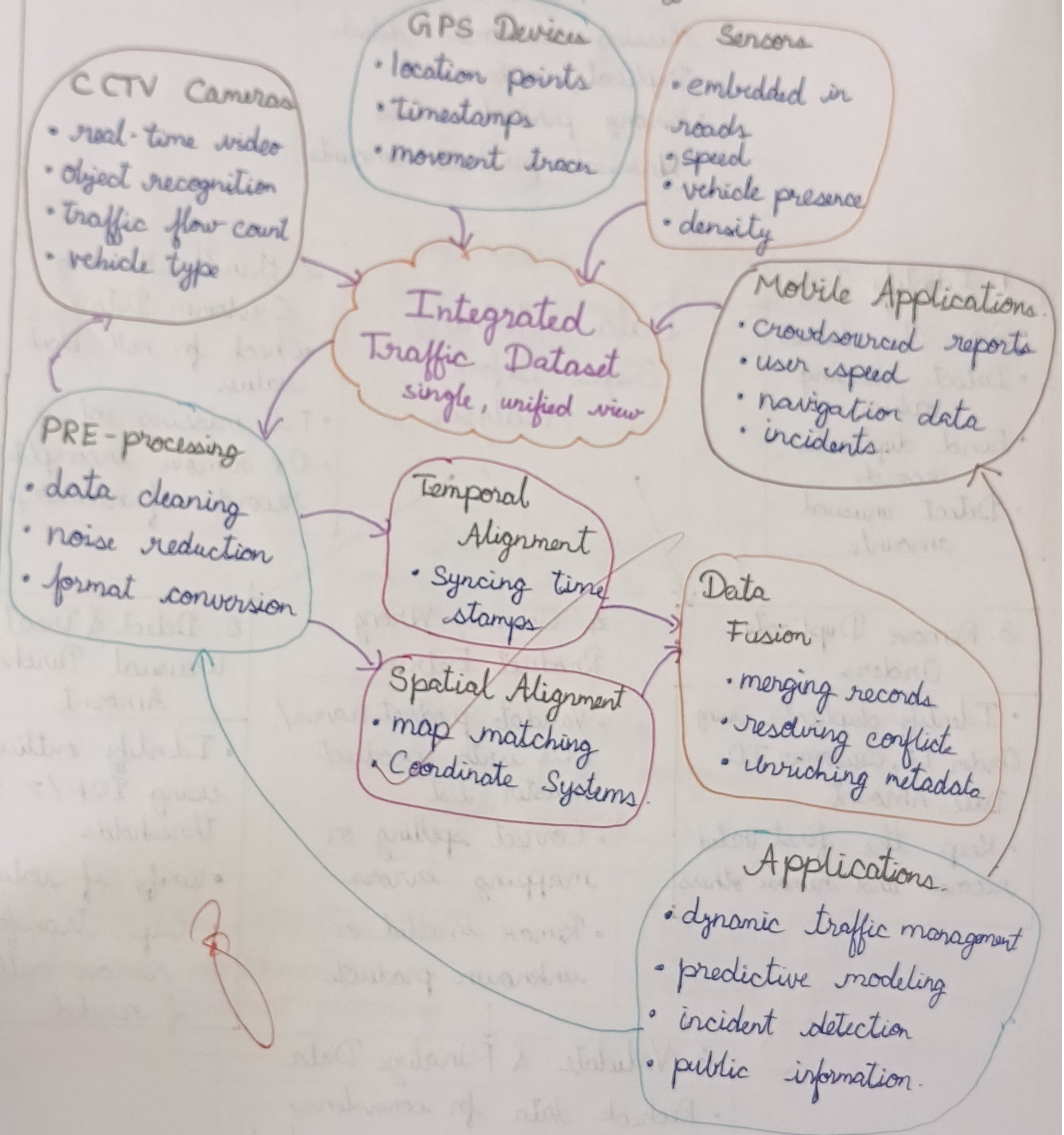
- Identify outliers using IQR/Z-Score/thresholds.
- Verify if valid.
- Cap, transform or remove outliers if needed.

6. Validate & Finalize Data

- Recheck data for consistency
- Ensure all issues are resolved
- Finalize clean dataset for analysis.

4. A smart city project collects traffic data from CCTV cameras, GPS devices, sensors, and mobile applications. Prepare a concept map showing how data from multiple sources can be integrated into a single dataset.

Integrating Smart City Traffic Data



A bank wants to prepare customer data for loan approval prediction. The dataset contains income, age, job type, loan amount, and credit score. Draw a concept map showing suitable data transformation techniques such as encoding, normalization, and feature scaling.